

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Pipeline Processing

Lecture No.- 4



By- Vishvadeep Gothi sir



Recap of Previous Lecture



Topic

Instruction Pipeline

Topic

Pipeline Hazards

Topics to be Covered



Topic

Data Hazards

Topic

Operand Forwarding

Topic

Branch Prediction

#Q. Consider a program which contains 200 instructions I1, I2 , I3 I200. Further consider a 5-stage pipeline with stages as: Instruction Fetch, Decode, Operand Fetch, Execution and Write-Back. The program contains only 1 branch instruction which is instruction I8 and its target is instruction I108. If during the execution of the program the branch is not taken then number of cycles required to execute this program in the given pipeline is 207 ?

$$n = 200$$

$$\begin{aligned} \text{w/o hazard} &= 5 + 200 - 1 = 204 \\ \text{stalls} &= 4 - 1 = 3 \\ \hline \text{Total} &= 207 \end{aligned}$$

hw. Questⁿ

I10 $\rightarrow$ I36

I54 $\rightarrow$ I133

k=4

Total I1 to I200

Target $\Rightarrow$ 2nd stage

Solⁿ

no. of inst^{ns} executed :-

I1 to I10 $\Rightarrow$ = 10

I36 to I54 $\Rightarrow$ $54 - 36 + 1 = 19$

I133 to I200 $\Rightarrow$ $200 - 133 + 1 = 68$

Total = 97

w/o branch hazard = $4 + 97 - 1 = 100$
stalls due to branch = $2 * (2 - 1) = 2$

Ans = 102



Topic : Data Hazard or Data Dependency

Result of an instruction is used as input in next

$R1 \leftarrow R2 + R3$
 $R6 \leftarrow R1 * R5$

$R1 = 15$
 $R2 = 6$
 $R3 = 9$
 $R5 = 8$

IF ID OF EX WB

IF ID — — OF EX WB

↓
stalls

↓
fetch of R1
should be after
WB of above instⁿ.

solⁿ of data hazard

s/w solution

(by compiler)

↓
Delayed Load

$R1 \leftarrow R2 + R3$

Instⁿ 1

Instⁿ 2

$R6 \leftarrow R1 * R5$

(pipeline cannot detect data dependency)

1	2	3	4	5	6
IF	ID	OF	EX	WB	
	IF	ID	OF	EX	WB
		IF	ID	OF	EX WB
			IF	ID	OF EX WB

H/w solution

→ H/w interlocking
→ operand forwarding (bypassing)

Compiler inserts independent instⁿs or no-operation instⁿs before dependent instⁿ to delay its operand fetch.

h/w interlock:-

providing stalls in execution (only if needed)

$R1 \leftarrow R2 + R3$

$R9 \leftarrow R8 + 1$

$R6 \leftarrow R1 * R5$

IF ID OF EX WB

IF ID OF EX WB

IF ID - OF EX WB

no. of stalls due to data dependency = (WB phase) - (OF phase)
in 2 immediate inst^{ns}

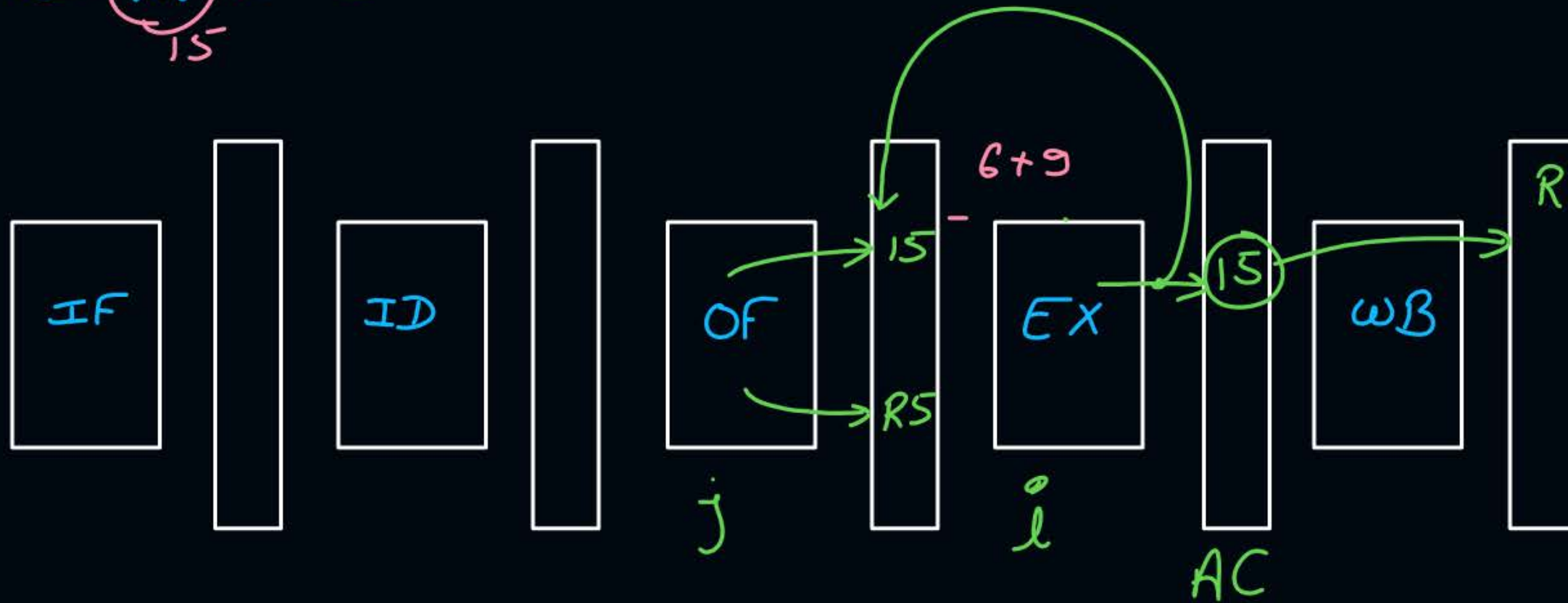
operand forwarding: (bypassing)

$$R2 = 6$$
$$R3 = 9$$

i: If ID of EX WB
j: If ID of OF EX WB

i: $R1 \leftarrow R2 + R3$

j: $R6 \leftarrow R1 * R5$



operand forwarding can save all stalls for only ALU to ALU dependencies.

Load instⁿ:-

$R1 \leftarrow m[Add.]$

IF ID OF EX WB

$R6 \leftarrow R1 * R5$

IF ID - - OF EX WB

store instⁿ:-

$R1 \leftarrow R2 + R3$

IF ID OF EX WB

$m[Add.] \leftarrow R1$

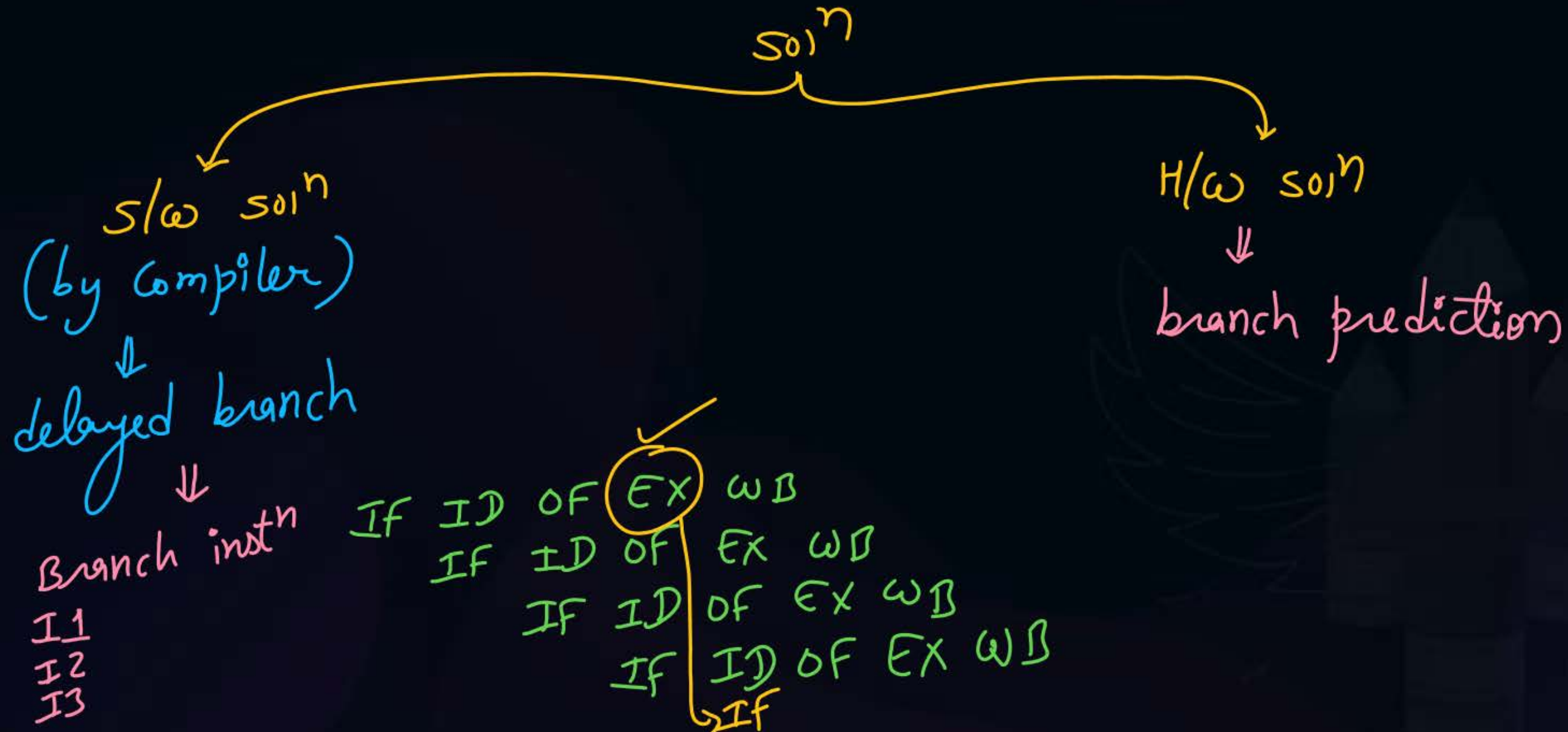
IF ID - - OF EX WB

even if operand forwarding is used then stalls are needed.



Topic : Control Hazard or Branch Difficulty

Hazards because of branch instructions



Branch prediction :-

Branch instⁿ IF (ID) OF EX WB

next instⁿ

IF ID - - - -

IF - - - -

IF _____

correct prediction $\Rightarrow$ save cycles

wrong - || - $\Rightarrow$ stalls

#Q. Consider a pipelined processor with the following four stages:

1. IF: Instruction Fetch
2. ID: Instruction Decode and Operand Fetch
3. EX: Execute
4. WB: Write Back

The IF, ID, and WB stages take one clock cycle each to complete the operation. The number of clock cycles for the EX stage depends on the instruction. The ADD and SUB instructions need 1 clock cycle and the MUL instruction needs 3 clock cycle in the EX stage. Operand forwarding is used in the pipelined processor. What is the number of clock cycles taken to complete the following sequence of instructions?

ADD R2, R1, R0

MUL R4, R3, R2

SUB R6, R5, R4

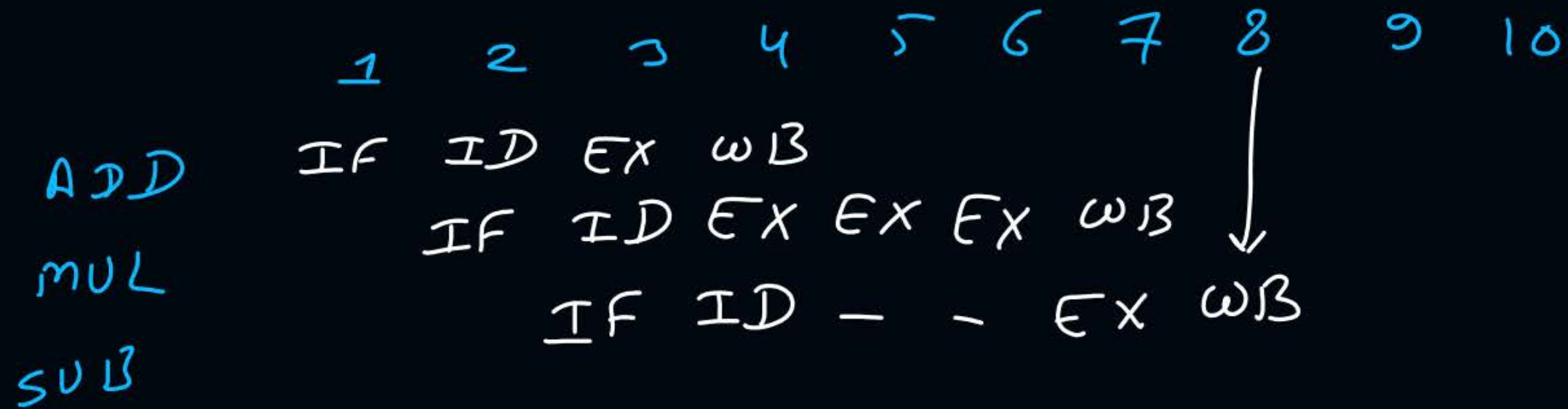
$R2 \leftarrow R1 + R0$

$R4 \leftarrow R3 * R2$

$R6 \leftarrow R5 - R4$

IF	_____	1			
ID	_____	1			
EX	_____		1	1	3
WB	_____	1			
			<u>stalls</u>		<u>2</u>

no stalls for data hazard
(operand forw.)
and also for branch hazard



[Solution]



w/o stalls no. of cycles = $k+n-1 = 4+3-1=6$

stalls for structural hazard = $1*2 = 2$

Total = 8

$$n = 7 + 4 + 5 = 16$$

#Q. In previous question if there are 7 ADD, 4 SUB and 5 MUL instructions then What is the number of clock cycles taken to complete such sequence of instructions?

$$\begin{array}{lcl} \text{w/o hazards, no. of cycles} & = & 4 + 16 - 1 = 19 \\ \text{stalls for structural hazard} & = & 2 * 5 = 10 \\ \hline \text{Total} & = & 29 \end{array}$$

#Q. ^{→ 3 instⁿ questⁿ} In previous question if no operand forwarding is ^{not} used what is the number of clock cycles taken to complete the given sequence of instructions?

ADD R2, R1, R0

MUL R4, R3, R2

SUB R6, R5, R4

$R2 \leftarrow R1 + R0$

$R4 \leftarrow R3 * R2$

$R6 \leftarrow R5 - R4$

no. of stalls for each data
dependency = $4 - 2 = 2$

$w/o \text{ hazard} = 4 + 3 - 1 = 6$
 $\text{stalls for structural hazard} = 2$
 $\text{data hazard} = 2 * 2 = 4$

Total = 12 Ans

Ques) In prev. questⁿ performance gain of using operand forwarding as compared to not using it?

Solⁿ

$$= \frac{\text{time w/o operand forwarding}}{\text{time with operand forwarding}}$$

$$= \frac{12}{8}$$

$$= 1.5$$

#Q. The instruction pipeline of a RISC processor has the following stages: Instruction Fetch (IF), Instruction Decode (ID), Operand Fetch (OF), Perform Operation (PO) and Writeback (WB). The IF, ID, OF and WB stages take 1 clock cycle each for every instruction. Consider a sequence of 100 instructions. In the PO stage, 40 instructions take 3 clock cycles each, 35 instructions take 2 clock cycles each, and the remaining 25 instructions take 1 clock cycle each. Assume that there are no data hazards and no control hazards.

The number of clock cycles required for completion of execution of the sequence of instruction is _____?



2 mins Summary



Topic

Data Hazards

Topic

Operand Forwarding

Topic

Branch Prediction



Happy Learning

THANK - YOU

